

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 5

2024

1. Let (X, d) be a metric space, and $A \subseteq X$. We have metric subspace (A, d_A) and the following:
 - A subset F of A is a closed subset of A iff $F = K \cap A$, for some closed subset K of X
 - A subset G of A is an open subset of A iff $G = H \cap A$, for some open subset H of X
 - $\text{cl}_A(B) = \text{cl}_X(B) \cap A$ for any subset B of A .

So we might wonder whether $\text{int}_A(B) = \text{int}_X(B) \cap A$, for any subset B of A .

In this question, the criterion for interior points given by Proposition 5.14(1) is most convenient. It says:

$$\begin{aligned} z \in \text{int}_X(B) &\iff \text{there exists } r > 0 \text{ with } B_X(z; r) = \{x \in X \mid d(z, x) < r\} \subseteq B \\ z \in \text{int}_A(B) &\iff \text{there exists } r > 0 \text{ with } B_A(z; r) = \{a \in A \mid d(z, a) < r\} \subseteq B. \end{aligned}$$

- (a) One of the inclusions $\text{int}_X(B) \cap A \subseteq \text{int}_A(B)$ and $\text{int}_A(B) \subseteq \text{int}_X(B) \cap A$ is always correct. Which one? Prove your answer.
 - (b) Find a counter-example for the other. As usual with counter-examples, you must specify both the set X and the metric d you are working with (as well as any other details required by the question).
2. Let (X, d) be a metric space, and $A \subseteq X$. Read the following proof and answer the questions below. (This is Proposition 5.18 in the notes, but with a different proof.)

Theorem Let (X, d) be a metric space, and $A \subseteq X$. We have metric subspace (A, d_A) .

A subset G of A is an open subset of A iff $G = H \cap A$, for some open subset H of X .

Proof: Let C' denote the complement of C as usual. (1)

Let $G \subseteq A$. (2)

Then G is open in A iff (3)

$A \cap G'$ is closed in A iff (4)

$A \cap G' = A \cap K$, for some closed subset K of X iff (5)

$A \cap G = A \cap K'$, for some closed subset K of X iff (6)

$G = A \cap H$, for some open subset H of X . (7)

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- (a) Why are lines (3) and (4) equivalent?
 - (b) Why are lines (4) and (5) equivalent?
 - (c) Prove the equivalence of the statements in lines (5) and (6).
 - (d) Why are lines (6) and (7) equivalent?

3. Let (X, d) be a metric space. Prove or disprove the following statements.

- (a) If A and B are subsets of X with $A \subseteq B$, then $\text{int}(A) \subseteq \text{int}(B)$.
- (b) If A is a subset of X , then $\text{int}(\text{bd}(A)) = \emptyset$.

End of syllabus for Test 1

4. **Theorem (A sufficient condition for two metrics to be equivalent)** Two metrics d and ρ on a set X are equivalent if there exist real numbers $m > 0$ and $M > 0$ such that, for all $x, y \in X$,

$$m d(x, y) \leq \rho(x, y) \leq M d(x, y).$$

To prove this, follow these steps:

- (a) Suppose $x_n \rightarrow x$ with respect to d . Use the inequality

$$0 \leq \rho(x, y) \leq M d(x, y)$$

to deduce that $x_n \rightarrow x$ with respect to ρ also.

- (b) Suppose $x_n \rightarrow x$ with respect to ρ . Use the inequality

$$0 \leq d(x, y) \leq \frac{1}{m} \rho(x, y)$$

to deduce that $x_n \rightarrow x$ with respect to d also.

5. (a) Let ρ be the usual metric on the reals, restricted to the set:

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Is ρ the discrete metric on S ? Is ρ equivalent to the discrete metric on S ?

- (b) Suppose that d_1 and d_2 are equivalent metrics on a set X , and that d_1 is an ultrametric. Does this mean that d_2 is an ultrametric as well?

6. Let X be a non-empty set. Consider

$$\mathcal{T} = \{A \subseteq X \mid A' \text{ is finite}\} \cup \{\emptyset\}$$

- (a) Show that $\mathcal{T}$ is a topology on X .
- (b) If X is a finite set this topology $\mathcal{T}$ is induced by a metric. Find such a metric.
- (c) Show that if X is an infinite set then this topology $\mathcal{T}$ is not induced by a metric.

7. Optional Challenge Question. The point of this question is to show that the sufficient condition of Question 4 above is not necessary.

Given a metric d on a set X , we define

$$d^*(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

In assignment 1 you have checked that d^* is a metric.

- (a) Prove that d and d^* are equivalent metrics.

[Hint: show that $B_d(x, r) \subseteq B_{d^*}(x, r)$ and $B_{d^*}(x, s) \subseteq B_d(x, r)$ for $s = \frac{r}{1+r}$.]

- (b) Prove that there is no $m > 0$ such that, for all $x, y \in \mathbb{R}$, $m d(x, y) \leq d^*(x, y)$ where d is the usual metric on the reals.